WORKED SOLUTION

STEP I 1996, Question 3

Pure • Divisibility of Integers

STEP I 1996, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q3

The key idea

Divisibility by a composite number such as 24 is easiest to handle one prime power at a time: $24 = 3 \times 8$ with 3 and 8 coprime, so we show divisibility by 3 and by 8 separately. Factorising gives us products of consecutive integers, and among any k consecutive integers one is a multiple of k. The other two parts yield to the same habit: factorise, or write n in a form that shows its remainder.

Solution

Part (i)

Taking out the common factor and using the difference of two squares,

$$n^5 - n^3 = n^3(n^2 - 1) = n^3(n-1)(n+1) = n^2\,(n-1)\,n\,(n+1).$$

Divisibility by 3. The numbers $n-1$, n, $n+1$ are three consecutive integers, so exactly one of them is a multiple of 3. Hence $3 \mid (n-1)n(n+1)$, and so $3 \mid n^5 - n^3$.

Divisibility by 8. We split into cases on the parity of n.

- If n is even, say $n = 2m$, then $n^3 = 8m^3$, which is a multiple of 8.
- If n is odd, then $n-1$ and $n+1$ are consecutive even numbers. One of any two consecutive even numbers is a multiple of 4, and the other is a multiple of 2, so $8 \mid (n-1)(n+1)$.

In either case $8 \mid n^3(n-1)(n+1) = n^5 - n^3$.

Ask yourself Why does knowing $3 \mid N$ and $8 \mid N$ give $24 \mid N$? Would $4 \mid N$ and $6 \mid N$ give the same?

Since 3 and 8 have no common factor, $8 \mid N$ means $N = 8a$, and then $3 \mid 8a$ forces $3 \mid a$ (as 3 is prime and doesn't divide 8). So $N = 24b$. The argument fails for 4 and 6, which

share the factor 2: the number 12 is divisible by both but not by 24.

Therefore $n^5 - n^3$ is divisible by 24 for every positive integer n, as required.

Part (ii)

Write $2^{2n} = 4^n$. Using the factorisation $x^n - 1 = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1)$ with $x = 4$,

$$2^{2n} - 1 = 4^n - 1 = (4-1)\left(4^{n-1} + 4^{n-2} + \cdots + 4 + 1\right) = 3\left(4^{n-1} + \cdots + 1\right).$$

The bracket is an integer, so $2^{2n} - 1$ is divisible by 3.

Equivalently, in modular arithmetic: $4 \equiv 1 \pmod 3$, so $4^n \equiv 1^n = 1 \pmod 3$. A proof by induction also works: if $4^k - 1 = 3a$, then $4^{k+1} - 1 = 4(4^k - 1) + 3 = 3(4a + 1)$.

Part (iii)

Since $3 \mid n - 1$, write $n = 3k + 1$ for some integer $k \geq 0$. Factorise

$$n^3 - 1 = (n-1)(n^2 + n + 1).$$

The first factor is $n - 1 = 3k$. For the second,

$$\begin{aligned} n^2 + n + 1 &= (9k^2 + 6k + 1) + (3k + 1) + 1 \\ &= 9k^2 + 9k + 3 \\ &= 3(3k^2 + 3k + 1). \end{aligned}$$

So each factor contributes a 3:

$$n^3 - 1 = 3k \cdot 3(3k^2 + 3k + 1) = 9k(3k^2 + 3k + 1),$$

which is divisible by 9, as required. (When $n = 1$ we get $k = 0$ and $n^3 - 1 = 0$, which is divisible by 9 too.)

The point is that $n - 1$ supplies only one factor of 3; the second comes from $n^2 + n + 1$. A neat way to see that second factor is $n^2 + n + 1 = (n-1)^2 + 3n$, a sum of two multiples of 3 whenever $3 \mid n - 1$.

Checks

- Part (i): $n = 2$ gives $32 - 8 = 24$ and $n = 3$ gives $243 - 27 = 216 = 9 \times 24$. Since $n = 2$ gives exactly 24, no larger number divides every $n^5 - n^3$, so 24 is the best possible.
- Part (ii): $n = 1, 2, 3$ give 3, 15, 63.
- Part (iii): $n = 4$ gives $63 = 7 \times 9$ and $n = 7$ gives $342 = 38 \times 9$, matching $9k(3k^2 + 3k + 1)$ with $k = 1$ and $k = 2$.

We checked all three results by computer for every n below 5000 (and for $n < 2000$ in part (ii)), and confirmed that the highest common factor of the values of $n^5 - n^3$ is exactly 24.

Common slips

- Arguing "$(n-1)n(n+1)$ is divisible by 6, and n^2 is …" and losing track of where the factor 8 comes from. The even case needs the n^3, not just n.
- Concluding divisibility by 24 from divisibility by 4 and by 6; the factors must be coprime.
- In part (iii), stopping at "$n-1$ is divisible by 3, so $n^3 - 1 = (n-1)(\dots)$ is divisible by 3", which gives only one factor of 3.

Beyond STEP

All three parts are really statements in modular arithmetic: the integers modulo m form a ring, and part (iii) says that if $n \equiv 1 \pmod 3$ then $n^3 \equiv 1 \pmod 9$. This is the first step of a general phenomenon called *lifting the exponent*: if $p \mid n-1$ then $p^{j+1} \mid n^{p^j} - 1$. Fermat's little theorem ($n^p \equiv n \pmod p$) explains part (i) too, since it gives $3 \mid n^3 - n$ directly. Davenport's *The Higher Arithmetic* is a beautiful introduction to these ideas.

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